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Practical example

Product								
ID	Building	Year of sale	Month of sale	Type of property	Property #	Area (ft.)	Price	Status
1030	1	2005	11	Apartment	30	743.09	$246,172.68	Sold
1029	1	2005	10	Apartment	29	756.21	$246,331.90	Sold
2002	2	2007	7	Apartment	2	587.28	$209,280.91	Sold
2031	2	2007	12	Apartment	31	1604.75	$452,667.01	Sold
1049	1	2004	11	Apartment	49	1375.45	$467,083.31	Sold
3011	3	2007	9	Apartment	11	675.19	$203,491.85	Sold
3026	3	2007	9	Apartment	26	670.89	$212,520.83	Sold
3023	3	2008	1	Apartment	23	720.81	$198,591.85	Sold
1031	1	2006	6	Apartment	31	782.25	$265,467.68	Sold
4023	4	2006	3	Apartment	23	794.52	$235,633.26	Sold
1036	1	2004	10	Apartment	36	1160.36	$317,473.86	Sold
1046	1	2006	8	Apartment	46	1942.50	$503,790.23	Sold
4035	4	2007	10	Apartment	35	794.52	$217,786.38	Sold
2036	2	2006	11	Apartment	36	1109.25	$460,001.26	Sold
2056	2	2007	4	Apartment	56	1400.95	$460,001.26	Sold
1047	1	2007	12	Apartment	47	1479.72	$448,134.27	Sold
5051	5	2006	3	Apartment	51	790.54	$249,591.99	Sold
2007	2	2006	8	Apartment	7	723.93	$196,142.19	Sold
3020	3	2007	4	Apartment	20	781.07	$258,572.48	Sold
3029	3	2007	4	Apartment	29	1127.76	$310,831.21	Sold
3015	3	2006	10	Apartment	15	720.70	$207,281.59	Sold
2004	2	2006	12	Apartment	4	649.69	$168,834.04	Sold
2006	2	2006	12	Apartment	6	1307.45	$396,973.83	Sold
5013	5	2007	9	Apartment	13	618.38	$188,743.11	Sold
1026	1	2005	3	Apartment	26	625.80	$179,674.08	Sold
2054	2	2006	6	Apartment	54	1203.29	$306,363.64	Sold
3033	3	2007	9	Apartment	33	670.89	$200,300.63	Sold
1025	1	2005	3	Apartment	25	1434.09	$382,041.13	Sold
3027	3	2006	8	Apartment	27	781.07	$245,572.79	Sold
3031	3	2007	3	Apartment	31	1596.35	$407,214.29	Sold
2043	2	2007	4	Apartment	43	1110.32	$355,073.40	Sold
3034	3	2007	4	Apartment	34	781.07	$256,821.64	Sold
3016	3	2007	4	Apartment	16	697.89	$226,342.80	Sold
1018	1	2004	10	Apartment	18	625.80	$191,389.87	Sold
2050	2	2006	9	Apartment	50	957.53	$297,008.97	Sold

Customer											
Customer ID	Entity	Name	Surname	Age at time of purchase	Interval Y	Gender	Country	State	Purpose	Deal satisfaction	Mortgage	Source
C0028	Individual	Madalyn	Mercer	19	18-25 1986	F	USA	California	Home	5	No	Website
C0027	Individual	Lara	Carrillo	22	18-25 1983	F	USA	California	Home	5	No	Website
C0112	Individual	Donavan	Flowers	22	18-25 1985	M	USA	California	Home	1	Yes	Client
C0160	Individual	Darien	Dorsey	22	18-25 1985	M	USA	California	Investment	3	Yes	Website
C0014	Individual	Alessandra	Perry	25	18-25 1979	F	USA	California	Home	4	No	Agency
C0125	Individual	Kaitlin	Owen	26	26-35 1981	F	USA	Virginia	Investment	5	No	Client
C0125	Individual	Kaitlin	Owen	26	26-35 1981	F	USA	Virginia	Investment	5	No	Agency
C0166	Individual	Terry	Forbes	26	26-35 1982	M	USA	California	Home	5	No	Client
C0034	Individual	Kole	Shannon	27	26-35 1979	M	USA	Arizona	Home	2	Yes	Website
C0170	Individual	Emmy	Singh	27	26-35 1979	F	USA	Virginia	Investment	3	Yes	Agency
C0009	Individual	Arabella	Ferrell	28	26-35 1976	F	USA	Oregon	Home	1	No	Agency
C0041	Individual	Christian	Costa	26	26-35 1980	M	USA	California	Home	5	No	Website
C0057	Individual	Michelle	Cameron	29	26-35 1978	F	USA	Nevada	Home	3	Yes	Website
C0061	Individual	Enrique	Cardenas	29	26-35 1977	M	USA	California	Home	2	No	Website
C0089	Individual	Amanda	Simon	29	26-35 1978	F	USA	California	Home	5	No	Agency
C0159	Individual	Kamden	Stewart	29	26-35 1978	M	USA	California	Home	5	No	Website
C0171	Individual	Skylar	Buchanan	29	26-35 1977	M	USA	Nevada	Home	4	Yes	Website
C0042	Individual	Michael	Juarez	30	26-35 1976	M	USA	Colorado	Home	4	Yes	Agency
C0093	Individual	Antonio	Porter	30	26-35 1977	M	USA	Arizona	Home	5	No	Client
C0093	Individual	Antonio	Porter	30	26-35 1977	M	USA	Arizona	Home	3	No	Client
C0051	Individual	Conner	Huff	31	26-35 1975	M	USA	Nevada	Home	5	No	Website
C0064	Individual	Joaquin	Mullins	31	26-35 1975	M	USA	California	Investment	5	No	Agency
C0064	Individual	Joaquin	Mullins	31	26-35 1975	M	USA	California	Investment	5	No	Agency
C0128	Individual	Kyla	Walker	31	26-35 1976	F	USA	Colorado	Home	4	Yes	Agency
C0019	Individual	Victor	Jensen	32	26-35 1973	M	USA	California	Home	4	No	Website
C0037	Individual	Tyler	Carr	32	26-35 1974	M	USA	California	Home	5	Yes	Client
C0127	Individual	Maia	Chandler	32	26-35 1975	F	USA	Utah	Home	1	Yes	Website
C0018	Individual	Rodrigo	Ramirez	33	26-35 1972	M	USA	California	Home	1	No	Agency
C0040	Individual	Gianni	Fritz	33	26-35 1973	M	USA	California	Home	3	No	Website
C0080	Individual	Janelle	Espinoza	33	26-35 1974	F	USA	California	Investment	4	No	Client
C0083	Individual	Julio	Leonard	33	26-35 1974	M	USA	Nevada	Home	2	Yes	Website
C0085	Individual	Melanie	Holland	33	26-35 1974	F	USA	California	Investment	3	No	Client
C0091	Individual	Gordon	Brown	33	26-35 1974	M	UK		Investment	4	No	Agency
C0007	Individual	Jaelyn	Berger	34	26-35 1970	F	USA	California	Home	3	No	Website
C0048	Individual	Adriana	Shaffer	34	26-35 1972	F	USA	California	Investment	2	No	Agency

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Types of data and levels of measurement

Task 1: What are the types of data and the levels of measurement of the following variables: Cust ID, Mortgage, Year of sale.

Solution:

Variable	Type of data	Level of measurement	Comment
Cust ID	Categorical	Nominal	This variable has the same properties as ID.
Mortgage	Categorical	Nominal	This is a Binary variable. Like a Yes/No question or Gender.
Year of sale	Numerical, discrete	Interval	Year is a numerical variable. It is always discrete. The level of measurement is questionable, but we would treat it as interval, as the 0 year would be the time when the Big Bang happened. The current BC-AD calendar was arbitrary chosen (similarly to degrees Celsius and Fahrenheit).

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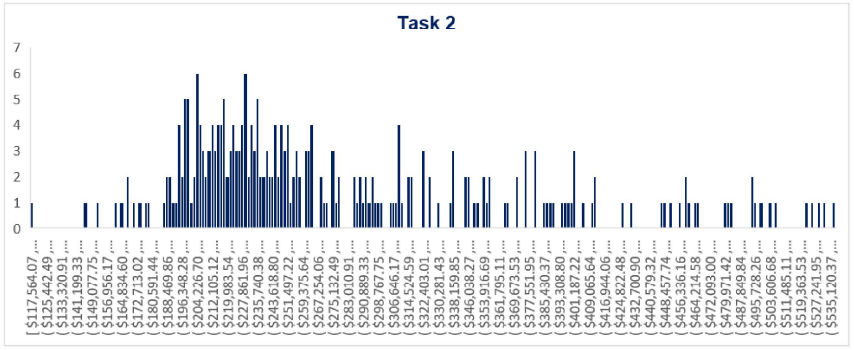
Histograms. Graphing numerical data

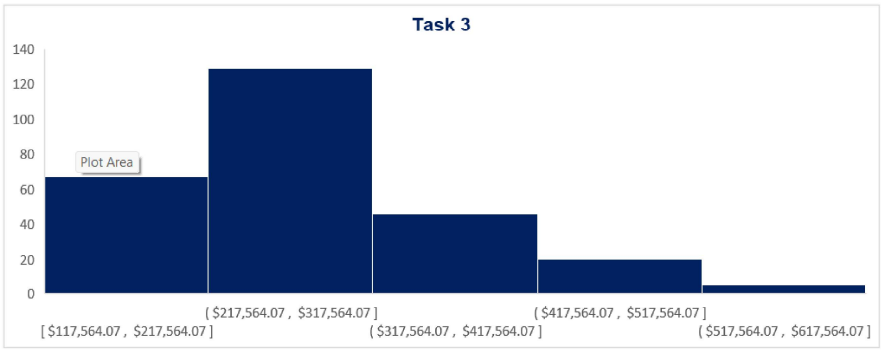
Task 2: Create a frequency distribution graph (that is a histogram with the highest possible number of bins - 267). Use data on all properties, no matter if sold or not.

Task 3: Create a histogram which represents the Price variable. Choose interval width (bins) of length $100,000. If you don't know how to do that, refer to the Course notes on descriptive statistics provided with the first lecture in this section.
Use the data on all properties, no matter if sold or not.

Task 4: Interpret the results.

Solution:





Task 4: The histograms point to similar insights - most of the properties' prices are concentrated in the interval ($217,564.07 to 317,564.07)

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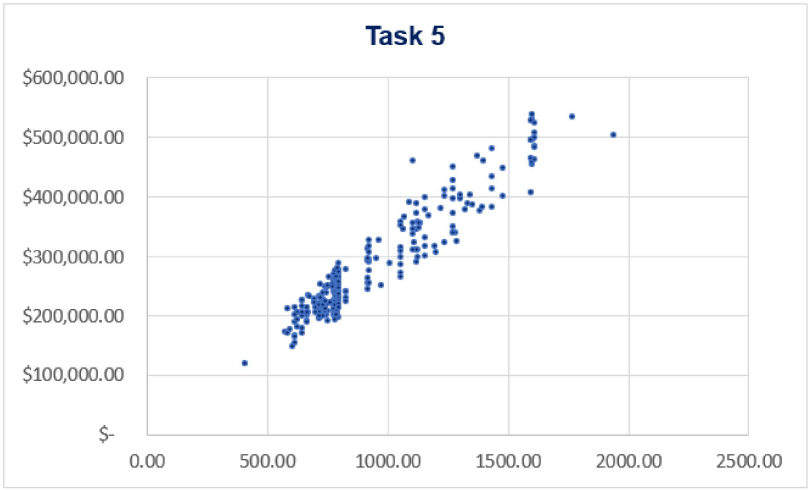
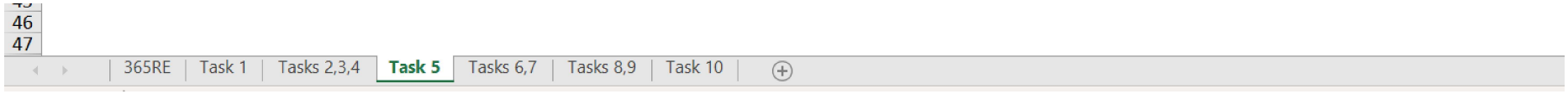
Scatter plot. Plotting numerical data; establishing relationship between variables.

Task 5: Create a scatter plot showing the relationship between Price and Area. Use the data on all properties, no matter if sold or not. Interpret the results.

Solution:



Task 5: The scatter plot shows a very strong linear relationship between Price and Area. This was to be expected as often RE companies price their property per square foot. Notice how for cheaper properties (lower areas respectively), the points are closer so the variance is smaller. The bigger the property, the bigger the difference in the price.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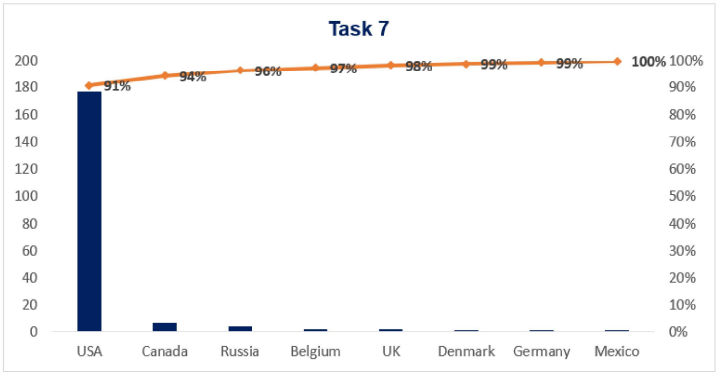
Frequency distribution table. Pareto diagram.

Task 6: Create a frequency distribution table, where you list all the countries from which the company has buyers. Count the absolute frequency, the relative frequency and the cumulative frequency.
Task 7: Create a Pareto diagram representing the data.

Solution:

Frequency distribution table

	Frequency	Relative frequency	Cumulative frequency
USA	177	91%	91%
Canada	7	4%	94%
Russia	4	2%	96%
Belgium	2	1%	97%
UK	2	1%	98%
Denmark	1	1%	99%
Germany	1	1%	99%
Mexico	1	1%	100%
Total	195	100%	



N.B. While a Pareto diagram is useless in this situation, it is still a good practice for you skills.

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Measures of central tendency, asymmetry and variability

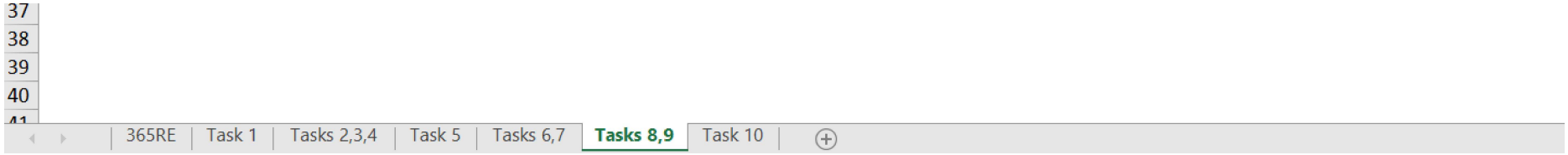
Task 8: Calculate the mean, median, mode, skewness, variance and standard deviation of Price for all properties, no matter if sold or not.
Task 9: Interpret the measures.

Solution:

Task 8:

Mean	$	281,171.90
Median	$	249,075.66
Mode	$	460,001.26
Skew		1.10
Variance		7942217700.92
St. dev.	$	89,119.12

Task 9: We will only comment on the skew, as it is a bit tougher. The skew is right (positive). This means that most properties are relatively cheap with a tiny portion that is more expensive.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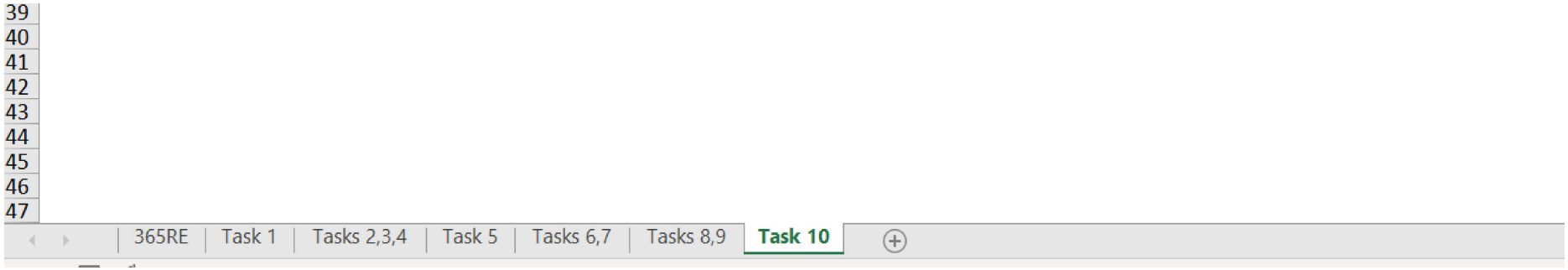
Measures of relationship between variables

Task 10: Calculate the covariance and correlation coefficient between Price and Area, no matter if the property is sold or not. Is the result in line with the scatter plot?

Solution:

Covariance	24147722
Correlation	0.9510874

Yes, the result is in line with the scatter plot. The two variables are greatly correlated.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